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Reconciliation systems and methods for unbounded streams

Abstract

A computing system is configured to (i) detect a window open sentinel indicating one or more reconciliation rules for a window within an unbounded stream of events, (ii) detect a shard open sentinel indicating one or more reconciliation rules for the shard, (iii) receive a set of events produced within the shard, (iv) detect a shard close sentinel indicating producer-generated reconciliation attribute(s) for the shard, (v) execute the reconciliation rules for the shard to thereby generate consumer-generated reconciliation attribute(s) for the shard, (vi) reconcile the consumer-generated reconciliation attributes for the shard against the producer-generated reconciliation attributes for the shard, (vii) detect a window close sentinel indicating producer-generated reconciliation attribute(s) for the window, (viii) execute the reconciliation rules for the window to thereby generate consumer-generated reconciliation attribute(s) for the window, and (ix) reconcile the consumer-generated reconciliation attribute(s) for the window against the producer-generated reconciliation attribute(s) for the window.

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Background/Summary

BACKGROUND

(1) Business organizations often rely on event-driven architectures to publish events that are subsequently consumed by unidentified downstream business applications. Business organizations may use such event-driven architectures to facilitate e-commerce transactions, social media websites, and the management of Internet of Things (IoT) devices. Many other uses of event-driven architectures are possible as well.

OVERVIEW

(2) Disclosed herein is new technology for implementing a reconciliation framework in which a producer computing system may convert an unbounded stream of events into a series of bounded, self-reconcilable streams of events that can be consumed and reconciled by a consumer computing

system.

(3) In one aspect, the disclosed technology may take the form of a method to be carried out by a computing system that involves (i) detecting a window open sentinel for a window within an unbounded stream of events, wherein the window open sentinel indicates one or more reconciliation rules for the window, (ii) detecting a shard open sentinel for a shard within the window, wherein the shard open sentinel indicates one or more reconciliation rules for the shard, (iii) receiving a set of events produced within the shard, (iv) detecting a shard close sentinel for the shard, wherein the shard close sentinel includes an indication of one or more producer-generated reconciliation attributes for the shard, (v) executing the one or more reconciliation rules for the shard for a set of events produced within the shard to thereby generate one or more consumer-generated reconciliation attributes for the shard, (vi) based on detecting the shard close sentinel, reconciling the determined one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard, (vii) detecting a window close sentinel, wherein the window close sentinel includes an indication of one or more producer-generated reconciliation attributes for the window, (viii) executing the one or more reconciliation rules for the window for a set of events produced within the window to thereby generate one or more consumer-generated reconciliation attributes for the window, and (ix) based on detecting the window close sentinel, reconciling the determined one or more consumer-generated reconciliation attributes for the window against the one or more producer-generated reconciliation attributes for the window.

(4) In some example embodiments, computing system is a consumer computing system and the method may also involve determining that the one or more consumer-generated reconciliation attributes for the shard match the one or more producer-generated reconciliation attributes for the shard, and transmitting, to a producer computing system, an indication that the shard was successfully reconciled. Additionally, or alternatively, the method may also involve determining that the one or more consumer-generated reconciliation attributes for the window match the one or more producer-generated reconciliation attributes for the window, and transmitting, to a producer computing system, an indication that the window was successfully reconciled.

(5) Further, in example embodiments, the computing system is a consumer computing system and the method may also involve determining that the one or more consumer-generated reconciliation attributes for the shard does not match the one or more producer-generated reconciliation attributes for the shard, discarding the set of events produced within the shard, and transmitting, to a producer computing system, a request to retransmit the set of events produced within the shard.

(6) Further yet, in example embodiments, the method may also involve creating a data object corresponding to the shard, and after each event in the set of events produced within the shard is received, adding the event to the data object for the shard.

(7) In such an embodiment, receiving the set of events produced within the shard may involve receiving at least one event in the set of events produced within the shard before detecting the shard open sentinel, and creating the data object corresponding to the shard may involve creating the data object corresponding to the shard based on receiving the at least one event in the set of events produced within the shard before detecting the shard open sentinel.

(8) Still further, in some example embodiments, executing the one or more reconciliation rules for the shard may involve executing the one or more reconciliation rules for the shard after all events in the set of events produced within the shard are received.

(9) Still further, in some example embodiments, executing the one or more reconciliation rules for the shard may involve executing the one or more reconciliation rules for the shard after each event in the set of events produced within the shard is received.

(10) Still further, in some example embodiments, the window open sentinel includes an indication of a window identifier, and each event in the set of events received within the window includes an indication of the window identifier.

- (11) Still further, in some example embodiments, the one or more reconciliation rules for the window may include instructions for determining (i) a given number of events produced within the window, (ii) a given number of shards produced within the window, and (iii) a signature for each event produced within the window.
- (12) Still further, in some example embodiments, the one or more reconciliation rules for the window may include instructions for determining one or more of (i) a summation of a given event attribute across all events produced within the window, (ii) a number of a given type of event produced within the window, or (iii) a signature for each of a given type of event produced within the window
- (13) Still further, in some example embodiments, the one or more reconciliation rules for the shard may include instructions for determining (i) a given number of events produced within the shard and (ii) a signature for each event produced within the shard.
- (14) Still further, in some example embodiments, the one or more reconciliation rules for the shard may include instructions for determining one or more of (i) a summation of a given event attribute across all events produced within the shard, (ii) a number of a given type of event produced within the shard, or (iii) a signature for each of a given type of event produced within the shard.
- (15) Still further, in some example embodiments, the one or more reconciliation rules for the window are different from the one or more reconciliation rules for the shard.
- (16) In yet another aspect, disclosed herein is a computing system that includes a network interface for communicating over at least one data network, at least one processor, at least one non-transitory computer-readable medium, and program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor to cause the computing system to carry out the functions disclosed herein, including but not limited to the functions of one or both of the foregoing methods.
- (17) In still another aspect, disclosed herein is a non-transitory computer-readable medium provisioned with program instructions that, when executed by at least one processor, cause a computing system to carry out the functions disclosed herein, including but not limited to the functions of one or both of the foregoing methods.
- (18) One of ordinary skill in the art will appreciate these as well as numerous other aspects in reading the following disclosure.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 depicts an example computing platform that employs an event-driven architecture.
- (2) FIG. 2 depicts a schematic example of an unbounded stream of events within an event-driven architecture.
- (3) FIG. 3 depicts an example unit of reconciliation within an unbounded stream of events in accordance with the disclosed technology.
- (4) FIG. 4 depicts a flow diagram of an example process carried out by a producer computing system in accordance with the disclosed technology.
- (5) FIG. 5 depicts a flow diagram of an example process carried out by a consumer computing system in accordance with the disclosed technology.
- (6) FIG. 6 is a simplified block diagram that illustrates some structural components that may be included in an example computing system.
- (7) Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following description, appended claims, and accompanying drawings, as listed below. The drawings are for the purpose of illustrating example embodiments, but those of ordinary skill in the art will understand that the technology disclosed herein is not

limited to the arrangements and/or instrumentality shown in the drawings.

DETAILED DESCRIPTION

(8) Disclosed herein is a new technology for converting an unbounded stream of events that may be generated in the context of an event-driven system architecture into a bounded, self-reconcilable stream of events. The disclosed technology may be carried out by any organization that engages in producing events, facilitating the delivery of events, and/or consuming events, examples of which include business organizations that facilitate e-commerce transactions, social media websites, and the management of Internet of Things (IoT) devices, among many other possibilities.

(9) Business organizations such as these typically employ one or more computing platforms that include, at least in part, an event-driven architecture for producing, facilitating the delivery of, and consuming events. A typical event-driven architecture is made up of a collection of discrete software components, sometimes referred to as “services,” each of which may carry out one or more functions within the event-driven architecture. These software components may be implemented using various different techniques. For example, the software components may be implemented in part using a microservices architecture where one or more of the software components are deployed in containers and communicate using respective APIs. As another example, one or more of the software components may utilize a serverless architecture where the given software component is implemented as a set of cloud-based functions that may only be run when another software component accesses its API. A computing platform implementing an event-driven architecture may integrate aspects of any of these various techniques, among others, in any combination.

(10) The physical instantiation of the one or more software components discussed above may also take various forms. In this regard, it should be noted that the physical hardware (e.g., servers, processors, communication interfaces, etc.) that performs the functions within the event-driven architecture might not be organized along the same logical lines as the associated software components. As one example, a given software component may be collectively implemented by two or more physically distinct computing systems within an overall computing platform. As another example, two or more logically separate software components (e.g., within separate containers or virtual machines) may be implemented using the same physical hardware. The software components may take other forms and may be implemented in other ways as well. Some of the structural components of the computing system(s) that might constitute an overall computing platform implementing an event-driven architecture are discussed further below in relation to FIG. 6.

(11) The events that are produced and consumed in an event-driven architecture may take various forms, one of which may include a change in state in the computing platform that implements the event-driven architecture. As one example, if the event-driven architecture facilitates e-commerce transactions, an event may include a purchase order, a product availability inquiry, or an item return, among other possibilities. As another example, if the event-driven architecture facilitates social media interactions across a social media platform, an event may include a notification that a user has posted an update (e.g., a message, a photo, a reaction to another user's post, etc.). As yet another example, if the event-driven architecture facilitates the management of IoT devices, an event may include a temperature sensor reading from a network-connected thermostat or a proximity alert from a network-connect motion detector, among other possibilities.

(12) As noted above, the discrete software components within an event-driven architecture may carry out various different functions. For instance, one or more of the discrete software components may generate events within the event-driven architecture. For purposes of the following discussion, the one or more software components (and the associated physical computing system(s)) that generate events within the event-driven architecture may generally be referred to as one or more “producers.” A given producer within the event-driven architecture may take various forms. As one example in the context of an e-commerce platform, the given producer may take the form of a

point-of-sale device, where the point-of-sale device generates an event for each sale or return. As another example, the given producer may take the form of a mobile application, where the mobile application generates an event in response to receiving an inquiry about the availability of an item for sale. As yet another example, the given producer may take the form of a retail web site, where the retail web site generates an event in response to a user placing a new order for an item. The given producer may take many other forms as well, depending on the application of the event-driven architecture.

(13) Further, one or more of the discrete software components discussed above may consume the generated events within the event-driven architecture. In this regard, consuming an event may generally refer to receiving the event, and perhaps performing some processing associated with the event as well. For purposes of the following discussion, the one or more discrete software components (and associated physical computing system(s)) that consume events within the event-driven architecture may be referred to as one or more “consumers.” A given consumer within the event-driven architecture may take various forms. As one example in the context of an e-commerce platform, the given consumer may take the form of a warehouse management database, where the warehouse management database may consume a produced sale event and in response, update its inventory and item availability records. As another example, the given consumer may take the form of a finance system, where the finance system may consume a produced sale event (e.g., the same sale event as the previous example) and in response, update financial records to reflect the sale. As yet another example, the given consumer may take the form of a customer relations system, where the customer relations system may consume an event corresponding to a question about availability of a particular item for sale and in response, generate a notification for a customer relations expert to respond to the inquiry. The given consumer may take many other forms as well, depending on the application of the event-driven architecture.

(14) Further still, one or more of the discrete software components discussed above may facilitate (i) the delivery of events from the one or more producers to the one or more consumers and (ii) the delivery of one or more responses from the one or more consumers to the one or more producers. These software components may collectively form a messaging system through which events and responses can be sent. Such a messaging system may take various forms. As one possibility, the messaging system may take the form of an event broker that receives, stores, and delivers events from a producer to one or more consumers. As another possibility, the messaging system may take the form of an event router that receives events from a producer and routes them to one or more consumers. As yet another possibility, the messaging system may take the form of a messaging bus that receives produced events from a producer and delivers the produced events to one or more consumers that subscribe to the producer. The messaging system may take other forms as well.

(15) Further, the messaging system may employ a variety of technologies to receive the events from a producer and deliver the events to a consumer. As one possibility, the messaging system may employ one or more queues, where the messaging system receives a produced event and places the received event in a queue that can be accessed by a consumer. As another possibility, the messaging system may employ one or more topics, where the messaging system applies additional metadata to the received event that facilitates the delivery of the event to consumers that have subscribed to the events corresponding to the topic. As yet another possibility, the messaging system may employ one or more logs, where the messaging system stores received events in a log that can be later accessed by a consumer. As still another possibility, the messaging system may employ streams, where the messaging system receives a produced event and makes the event available to the consumer in real time. The messaging system may employ other technologies as well.

(16) Turning to FIG. 1, an illustrative example of a computing platform **100** implementing an event-driven architecture is shown. The event-driven architecture includes one or more event producers, illustrated by way of example as producer **110a**, producer **110b**, and producer **110c**. The

event-driven architecture also includes a messaging system **120** and one or more event consumers, illustrated by way of example as consumer **130a**, consumer **130b**, and consumer **130c**. The event producers may be configured to produce events associated with the computing platform **100** and may take various forms. As one possibility, event producer **110a** may be a point-of-sale system that is configured to produce payment transaction events each time an item is purchased. Accordingly, each event may be a data object that includes information about the payment transaction. For instance, each event may be organized according to a schema that defines certain data fields (e.g., a globally unique event identifier, a merchant identity, a payment amount, a transaction date, a transaction time, etc.) and corresponding values for the included data fields.

(17) As another possibility, event producer **110b** may be a customer service system that is configured to produce an event each time a dispute related to a payment transaction event is initiated (e.g., each time a business or customer disputes a transaction). Accordingly, each event may be a data object that includes information about the disputed transaction. For instance, a disputed transaction event may include a schema that may define different data fields (e.g., the disputed payment event's globally unique event identifier, a merchant identity, a disputed payment amount, etc.) and corresponding values for those data fields. The event producers, and the associated events they produce, may take many other forms as well.

(18) Each of the producers shown in FIG. **1** may be configured to produce events and send them to the messaging system **120**. In this respect, messaging system **120** may function to receive, store, and deliver the produced events. For instance, the messaging system **120** may receive the produced events via a respective communication link that connects the messaging system **120** with each producer. The communication links may include any one or more of point-to-point links, Personal Area Networks (PANs), Local-Area Networks (LANs), Wide-Area Networks (WANs) such as the Internet or cellular networks, cloud networks, and/or operational technology (OT) networks, among other possibilities. Further, the communication networks and/or links that make up each respective communication link may be wireless, wired, or some combination thereof, and may carry data according to any of various different communication protocols. Messaging system **120** may receive the produced events from each of the producers in other ways as well.

(19) Messaging system **120** may store the produced events in various ways. As one example, messaging system **120** may employ a data store, where messaging system **120** stores each event in the data store as it is received. In some implementations, when messaging system **120** delivers a produced event to one or more consumers that are designated to receive the event, messaging system **120** may purge the event from the data store. In some other implementations, messaging system **120** may employ an event log, where messaging system **120** logs each event in the event log as it is received. In such an implementation, when messaging system **120** delivers a produced event to a consumer **130**, messaging system **120** retains the event in the event log. This may be beneficial in some situations, as it may allow the messaging system **120** to resend events if issues arise during delivery of the produced events. Messaging system **120** may store the produced events in other ways as well.

(20) Messaging system **120** may deliver the produced events in various ways. As one example, messaging system **120** may be configured to push the produced events to appropriate consumers. For instance, one or more consumers may subscribe to a given type of produced event (e.g., payment transaction events). Thereafter, when messaging system **120** receives an event of the given type, messaging system **120** pushes the given event to consumer **130a** and consumer **130b**. As another example, messaging system may publish received events in an event stream, which may be accessed by one or more consumers to retrieve the events. Messaging system **120** may deliver the produced events in other ways as well.

(21) Each of the consumers shown in FIG. **1** may be configured to consume produced events, which might include optionally performing one or more additional operations associated with the event, as noted above. As one example, consumer **130a** may be a pricing system that is configured

to consume payment transaction events (e.g., events related to purchases made using credit cards) that are produced by a point-of-sale system producer. Consumer **130a** may consume the payment transaction events and calculate a price to levy on each respective transaction (e.g., to be charged to the issuer of the credit card). As another example, consumer **130b** may be a dispute system that is configured to consume dispute events (e.g., events produced each time a business or customer disputes a transaction). Consumer **130b** may consume the events and calculate a fee to be levied on the dispute event (e.g., fees to be refunded). The event consumers may take other forms as well.

(22) It should be noted that, in the context of the computing platform **100** shown in FIG. **1** and other similar computing platforms that implement an event-driven architecture, there may not be a one-to-one correspondence between a producer and a consumer for each produced event. For instance, a single event that originates from a given producer might be consumer by two or more consumers. Similarly, a given consumer might consume events produced by many different producers, which may then be aggregated by the consumer before it carries out one or more operations on the aggregated event data. Various other examples are also possible.

(23) In some implementations, the various software components within an event-driven architecture may produce and consume events in the form of an “unbounded” stream of events. An unbounded stream of events is a continuous stream of events with no defined beginning or end. In the context of an event-driven architecture, an unbounded stream of events means that producers can continuously produce events without temporal boundaries (e.g., the producers produce events as they occur, and the consumers consume events as they are received). In this regard, an unbounded stream of events can be useful for delivering data within a computing platform in real time, or close to real time.

(24) FIG. **2** shows one illustrative example of an unbounded stream of events within an event-driven architecture. In FIG. **2**, a first software component associated with a first computing system, e.g., a producer **210**, produces and sends events to a second software component associated with a second computing system, e.g., a messaging system **260**, as part of an unbounded stream of events **200**. In FIG. **2**, event **220**, event **230**, event **240**, and event **250** are depicted, however the unbounded stream of events **200** may additionally include events both preceding event **220** and following event **250**. In some implementations, each event produced by producer **210** is immediately made available to messaging system **260**, which in turn may make the events immediately available to one or more downstream consumers. In some other implementations, producer **210** may generate multiple events before making them available to the messaging system **260**. In this regard, it should be understood that producer **210** may produce event **220**, event **230**, event **240**, and event **250** at any time. For instance, the four events shown in FIG. **2** might be produced approximately every 15 seconds over the course of one minute, or they might be produced at random times over the course of one hour or an entire day. Other examples are also possible.

(25) However, an unbounded stream of events in an event-driven architecture may be associated with various drawbacks. One potential drawback associated with unbounded streams of events is the difficulty in ensuring data reliability. For instance, because the events produced as part of an unbounded stream have no defined beginning or end, and can be produced at any time, it is difficult for a consumer to detect whether it is receiving every event that is produced. In some situations, an event that is produced by a producer may fail to reach the messaging system for any number of reasons (e.g., temporary network failure, third-party tampering). Similarly, an event may fail to be delivered from the messaging system to the consumer. In turn, this may expose the organization responsible for producing or consuming the event to operational or reputational risk.

(26) For similar reasons, it also may be difficult to ensure data continuity of an unbounded stream of events. For instance, a consumer may have no way to detect whether additional, extraneous events have been injected into the stream, or whether a given, legitimate event has been ejected in favor of an illegitimate event that may be injected in its place.

(27) Another potential drawback associated with unbounded streams of events in an event-driven architecture is the difficulty in ensuring data integrity—e.g., ensuring that the data within a received event has not been altered from the time the event was published by the producer. In this respect, an event may be exposed to tampering by a third-party after a producer generates the event. For instance, if a producer generates an event that is related to a financial transaction, a third-party may alter the values of the financial transaction at some point during intermediate processing or transit. As above, the consumer may have no way to detect such alterations, which may lead to the consumption and further processing of altered events.

(28) The drawbacks described above may be particularly applicable when the computing platform that implements an unbounded stream to deliver events is operated by a financial services institution, as problems with data reliability, continuity, and/or integrity may directly affect the processing of payment transactions, the collection of fees, and/or customer satisfaction. For instance, altered financial transactions may constitute fraudulent activity that leads to economic losses for one or more entities involved (e.g., consumers, merchants, credit card issuers, etc.). Further, if an event is generated that pertains to a disputed transaction and that event is never delivered to the appropriate consumer, the disputed transaction may not be resolved promptly, or at all, which in turn may result in poor customer satisfaction and further economic losses.

(29) For the reasons above, it may be advantageous to implement a framework that allows consumers to reconcile an unbounded stream of events from a producer in a way that improves a consumer's ability to verify reliable event delivery, continuity, and data integrity of the produced events. In this regard, “reconciliation” may refer to a process by which a consumer attempts to confirm both the number and identity of events it receives as well as certain attributes associated with the events.

(30) However, implementing a reconciliation framework in the context of an unbounded stream of events is challenging for various reasons. As discussed above, unbounded streams of events are not bounded by time or restricted to a particular order and thus it is difficult for a consumer to verify that an event was produced in the first place (and thus, that the event should be expected). In addition, it is difficult for a consumer to verify when a consumed event was produced and/or delivered in relation to other events, as the events that are produced in an unbounded stream may become reordered during intermediate processing or transit. Still further, it is difficult for a consumer to verify whether the data within a received event is the same as the data that was produced by the producer. Accordingly, there is a need to implement a reconciliation framework that can be used to verify event delivery, reliability, and integrity of events in an unbounded stream.

(31) In view of these shortcomings, a new technology is discussed herein for implementing a reconciliation framework in which a producer (e.g., a producer software component operating on a producer computing system) may convert an unbounded stream of events into a series of bounded, self-reconcilable streams of events that can be consumed and reconciled by a consumer (e.g., a consumer software component operating on a consumer computing system).

(32) In accordance with one aspect of the producer computing system, a given producer may function to inject boundary markers into an unbounded stream of events. These boundary markers are also referred to herein as “sentinels” herein and serve to divide the unbounded stream into one or more units that may be reconciled (e.g., a unit of reconciliation, or UoR) by a consumer. For instance, an unbounded stream of events may be divided into one or more windows that contain a given number of events. Each window may be further sub-divided into one or more shards that contain a corresponding number of events from within the window, and so on.

(33) At a high level, each sentinel that is injected into the unbounded stream of events by the producer may take the form of a data object that includes information signaling either the beginning (e.g., an “open” sentinel) or the end (e.g., a “close” sentinel) of a UoR. Further, the producer may include within each sentinel information that enables a consumer to perform reconciliation on the events within each UoR. The information in a given sentinel that enables

reconciliation by a consumer may take various forms.

(34) For example, in a given open sentinel (e.g., a window open sentinel or a shard open sentinel), the producer may include an identification of one or more reconciliation rules for the UoR that provide the basis for performing reconciliation on the events within the UoR. In general, the reconciliation rules of a given UoR may involve instructions for performing one or more operations associated with the events within the UoR, such as counting the number of events in the UoR or calculating a sum of the values within a given data field across all events in the UoR. Various other types of reconciliation rules are possible and will be discussed in further detail below.

(35) After producing each of the events in the UoR, the producer may execute the one or more reconciliation rules indicated in the open sentinel and thereby generate one or more producer-generated reconciliation attributes. In turn, the producer may include these one or more producer-generated reconciliation attributes in the corresponding close sentinel (e.g., a window close sentinel or a shard close sentinel) for the UoR, which may be injected into the unbounded stream after all events in the UoR are produced.

(36) From the consumer's perspective, the open sentinel provides an indication of the reconciliation rules that are to be used and the close sentinel includes an indication—in the form of the producer-generated reconciliation attributes—of the result that the consumer should obtain when executing the reconciliation rules. Accordingly, the consumer may execute the reconciliation rules in relation to the events it consumed within the UoR and thereby generate one or more consumer-generated reconciliation attributes. The consumer may then compare the consumer-generated reconciliation attributes with the producer-generated reconciliation attributes. If the attributes are the same, the consumer may determine that reconciliation of the UoR was successful. On the other hand, if one or more of the attributes differs, the consumer may determine that reconciliation was unsuccessful. Based on this determination, the consumer may transmit a request for the producer to re-send the events in the UoR (e.g., a replay request). Other examples are also possible and will be discussed in further detail below.

(37) Based on the foregoing, it will be appreciated that the reconciliation framework discussed herein results in an unbounded stream of events that is self-reconcilable based on information that is embedded in the stream itself. Accordingly, the producer may send the unbounded stream of events, including the injected sentinels, to a messaging system, which may deliver the stream to one or more consumers, without requiring any additional communications from the producer to the consumer(s) regarding the reconciliation of the stream (although the consumer may notify the producer of successful and/or unsuccessful reconciliation events, as discussed below). Further, any consumer of the stream may, after successful reconciliation of each UoR, become a producer of the same stream to further downstream consumers. In this regard, the events and data within the reproduced stream may be afforded the same level of trust as if the stream had come from the original producer.

(38) Still further, the self-reconcilable nature of the unbounded stream allows for reconciliation to be performed at any and all points through which the stream passes within an event-driven architecture. For example, in some implementations, the messaging system of the event-driven architecture may perform reconciliation of the unbounded stream, even though the messaging system may not undertake any further processing associated with the events. Thereafter, if reconciliation of a UoR within the stream fails at the consumer, the consumer may send a replay request to the messaging system, rather than the producer. Other implementations are also possible.

(39) Turning now to FIG. 3, an illustrative example is depicted in which a portion of an unbounded stream of events may be converted into a series of bounded, self-reconcilable streams of events via the reconciliation framework described herein. As shown in FIG. 3, a first software component associated with a first computing system, e.g., a producer **302**, produces and sends events that are eventually received by a second software component associated with a second computing system, e.g., a consumer **304**, as part of an unbounded stream of events **300**. In this regard, and consistent

with the discussion above, it should be understood that the unbounded stream of events **300** shown by way of example in FIG. **3** may be handled by one or more additional intermediate software components between the producer **302** and the consumer **304**, such as a messaging system that facilitates delivery of the events, as shown in FIG. **1** and FIG. **2**. However, in the interest of focusing on the operations of the producer **302** and the consumer **304** within the disclosed reconciliation framework, the messaging system is not depicted in FIG. **3**.

(40) At a high level, the example in FIG. **3** shows four events—event **314**, event **316**, event **324** and event **326**—that form a part of an unbounded stream of events **300**, into which the producer **302** has injected several sentinels to divide the stream into self-reconcilable units. In particular, the producer has injected a window open sentinel **308**, marking the beginning of a window **306**, and a window close sentinel **330**, marking the close of the window **306**. Further, the producer has injected respective open and close sentinels for two shards—shard **310** and shard **320**—that are sub-units of the window **306**. Accordingly, each shard contains a sub-portion of the events within the window, e.g., two events each. In this regard, it should be understood that FIG. **3** represents just one simplified example and that in some other implementations, a given window might contain more or fewer shards that in turn might contain more or fewer events, and, among other possibilities.

(41) Referring now to the window **306**, the producer **302** may initially determine the start of the window **306** within the unbounded stream of events **300**. Generally speaking, the producer **302** may determine the start of a new window following the end of a previous window in the unbounded stream. In turn, the producer **302** may determine the end of a given window in an unbounded stream based on various criteria, which will be discussed further below in relation to the window close sentinel **330**.

(42) Based on determining the start of window **306**, the producer **302** may generate a window open sentinel **308** that includes information to facilitate the eventual reconciliation of the window **306** by one or more downstream consumers, such as the consumer **304**. The information within the window open sentinel **308** to facilitate reconciliation may take various forms. For instance, the window open sentinel **308** may include metadata indicating a unique window identifier that the producer **302** may associate with each event that is produced within the window (e.g., by tagging the event with the window identifier). This may allow the consumer **304** to associate each received event with the correct window at the time of reconciliation.

(43) The information within the window open sentinel **308** may additionally include an identification of one or more reconciliation rules for performing reconciliation of the events within the window **306**. In this regard, the reconciliation rules may take the form of metadata that encodes instructions for performing one or more operations on the set of events that are included within the window **306**. The reconciliation rules may take various forms and may be defined by the producer **302** based on one or more organization needs.

(44) For example, one type of reconciliation rule may include an instruction to count the number of events that are included within the window **306**. As noted above, the producer **302** may execute this rule after all the events within the window **306** are published and may include the resultant value (e.g., as part of one or more producer-generated reconciliation attributes) in the window close sentinel **330**, as discussed further below. The consumer **304** may then use this value as the expected result when it executes the same rule for the events that it received in the window **306**. If the value the consumer **304** obtains after executing the rule is the same as the expected result, the consumer **304** may determine that reconciliation according to the rule was successful. Another, similar type of reconciliation rule may include an instruction to count the number of shards that are included within the window **306**.

(45) Although the reconciliation rules discussed above may enable a consumer **304** to validate that it consumed the correct number of events within a given window, they might not necessarily enable the consumer **304** to validate that the identity of the events it consumed were all the same as the

events that were produced by the producer **302**. For example, it may be possible for an event within window **306** to be ejected from the unbounded stream of events **300** and replaced by a substitute event (e.g., by a third party). In these situations, reconciliation rules that only include counting the number of events or shards within the window may not identify the replaced event.

(46) For this reason, the reconciliation rules may additionally or alternatively include an instruction to verify an event signature across all events in the window. In this regard, the events produced by the producer **302** may have a schema that includes a data field for a unique event identifier that may be used to generate an event signature. For instance, the reconciliation rule may include instructions for combining the unique event identifiers for each event using one or more cipher techniques. Accordingly, the value that results is based on each individual event identifier. In addition, the reconciliation rule may include instructions for generating a cryptographic hash of the resultant value using a hashing algorithm that is included as part of the reconciliation rule in the window open sentinel **308**. The cryptographic hash of the resultant value may then be included in the window close sentinel **330** (e.g., as a producer-generated reconciliation attribute). Thus, the consumer **304** can recalculate the hashed value using the hashing algorithm in the window open sentinel **308**, and then compare the resultant value to the value stored in the window close sentinel by the producer **302**.

(47) In some implementations, the producer **302** may define still further reconciliation rules that enable reconciliation of the events within the window **306** on a more granular level, such that the consumer **304** can validate the integrity of the information contained within the events. This may be useful in situations where it is desirable to prevent alterations to information within the events, where the alterations do not change the number or identity of the events within the window **306**, and thus may not be detected by the reconciliation rules noted above. For instance, a given event might be altered by a third party to change the value of a payment transaction amount (e.g., from \$50 to \$500).

(48) One example of this type of reconciliation rule may include instructions for aggregating (e.g., adding) the values from a given data field across all events within the window **306**. For instance, the producer **302** may produce payment transaction events, and the reconciliation rule may include instructions for aggregating the value from a “Payment Amount” field for each respective event within the window **306**. When the producer **302** executes the rule, the resultant value may be a total value (e.g., \$30,000) that is included in the window close sentinel **330** (e.g., as a producer-generated reconciliation attribute). Accordingly, the consumer **304** can execute the same rule at the time of reconciliation and validate that the aggregated value it obtains is the same as the producer's, from which it may be inferred that none of the “Payment Amount” values from the events within the window were altered. An aggregation rule may also take other forms (e.g., a mean aggregation rule) and may be used on other data fields as well.

(49) Another example of a more granular reconciliation rule may be a grouping rule that includes instructions for counting the occurrences of each unique value within a given data field across all events within the window **306**. For instance, the reconciliation rule may include instructions for grouping the values within a “Merchant” field within the payment transaction events of the prior example. Within the window **306**, there may be 100 total events including payment transactions that occurred at three different merchants (in this regard, the producer **302** might represent a computing system operated by an acquirer that handles payment transactions from multiple merchants). Accordingly, the producer **302** executes the reconciliation rule and generate resultant attributes indicating 30 transactions corresponding to the first merchant, 30 transactions corresponding to the second merchant, and 40 transactions corresponding to the third merchant. As above, these attributes may be included in the window close sentinel **330**, enabling the consumer **304** to validate during reconciliation that the distribution of events among the consumers is correct. A grouping rule of this type may be applied to other data fields as well, such as a “Dispute Type” field in the context of a dispute resolution platform, among other possibilities.

(50) Yet another example of a more granular reconciliation rule may involve instructions for generating a signature that incorporates the values from one or more data fields across all events within the window **306**, similar to the event signature rule discussed above. For instance, referring again to the unbounded stream of payment transaction events discussed in the prior examples, the reconciliation rule may include instruction for (i) concatenating the values of the respective “Payment Amount” and “Merchant” fields into a single value for each event (e.g., in the form “Payment Amount Merchant” (ii) generating a cryptographic hash of the combined value for each event, using a hashing algorithm included in the window open sentinel **308**, and (iii) combining the hashed values into a single hashed value using one or more cipher techniques. Similar to the examples above, the resultant value that the producer **302** generates when executing the reconciliation rule is dependent on every value, in both data fields, across all events in the window **306**. Thus, the consumer’s successful reconciliation of the window **306** may provide a strong indication that each of the values was unchanged. In this way, the producer **302** might define a signature rule that designates one or more data fields within the unbounded stream of events whose integrity is most important to validate during reconciliation. Other examples are also possible, including signature rules that involve more or fewer data fields.

(51) The one or more reconciliation rules may take other forms as well.

(52) As discussed throughout the examples above, the producer **302** may execute the one or more reconciliation rules included in the window open sentinel **308** after all of the events in the window **306** have been produced. In this regard, the producer might execute the reconciliation rules incrementally, updating a set of producer-generated reconciliation attributes after each and every event is produced. Alternatively, the producer **302** might execute the reconciliation rules for the entire window **306** after the last event (e.g., event **326**) is produced. The producer **302** may execute the reconciliation rules at other times as well, including combinations of the above.

(53) Regardless of when the producer **302** executes the reconciliation rules, it may include the resulting producer-generated reconciliation attributes in the window close sentinel **330**. In line with the examples above, the producer-generated reconciliation attributes may include a set of values, with one or more of the values corresponding to each of the reconciliation rules that were executed by the producer **302**. Each of these values, in turn, may represent an expected value that the consumer **304** may use as a reference to compare its own resultant values when executing the same reconciliation rules on the events that it consumed within the window.

(54) In some implementations, the producer **302** might not include the producer-generated reconciliation attributes directly in the window close sentinel **330**. Rather, the producer **302** may post the producer-generated reconciliation attributes to a secure registry or the like, and provide a link (e.g., a URI) or similar pointer in the window close sentinel that enables the consumer **304** to access the producer-generated reconciliation attributes. Other implementations are also possible for providing the consumer **304** with access to the producer-generated reconciliation attributes without including them directly in the window close sentinel **330**.

(55) It should be noted that, while one or more of the reconciliation rules above may be included in the window open sentinel **308** as discussed, it is also possible that the producer **302** and the consumer **304** may be configured to execute one or more of the above rules automatically, without reference to the window open sentinel **308**. For instance, both the producer **302** and the consumer **304** may be configured to count the number of events within the window **306**, and the window open sentinel **308** may lack any instruction for doing so. However, the producer **302** may still include the result of its execution of this automatic reconciliation rule in the window close sentinel **330**, as a reference for the consumer **304** to use during reconciliation. The producer **302** and the consumer **304** may be preconfigured to execute any of the reconciliation rules discussed above in this way.

(56) In some implementations, the producer **302** and the consumer **304** might exclusively execute reconciliation rules that they are already configured to execute, such that the window open sentinel

308 does not provide an indication of any additional reconciliation rules. In such an example, the open and close sentinels may nonetheless still provide for the division of the unbounded stream into reconcilable units, defining the boundaries for when the producer **302** and the consumer **304** should execute the pre-encoded reconciliation rules. Further, the window close sentinel **330** may still provide a vehicle for communicating the producer-generated reconciliation attributes to be used by the consumer **304**.

(57) In addition to an indication of the producer-generated reconciliation attributes, the window close sentinel **330** may include other information that may also facilitate reconciliation of the window **306**. For instance, the window close sentinel **330** may include an indication of the unique identifier for the window **306** that the producer **302** included in the window open sentinel **308**.

(58) The producer **302** may determine when to generate the window close sentinel **330** in various ways. As one example, the producer **302** may be configured to inject a window close sentinel after a prescribed number of events has been produced, such as 100 events, after which the producer **302** will inject a new window open sentinel and start the count again. As another example, the producer **302** might be configured to produce windows based on a prescribed time period, such as every 30 seconds. Accordingly, the producer **302** may generate and inject a window close sentinel 30 seconds after a window open sentinel was injected, following by another window open sentinel.

(59) Other examples are also possible, including combinations or variations of the above. For instance, in some situations, a producer **302** might produce events relatively slowly such that waiting for an event count to be met before sending a window close sentinel might undesirably delay reconciliation of some produced events. In these situations, the producer **302** might be configured to inject no window open sentinels into the stream until an event is produced, and then to inject a corresponding window close sentinel no later than 30 seconds (or some other time period) after the window open sentinel is injected. Various other possibilities also exist.

(60) The consumer **304** may consume published events within the window **306** as they are received and may generally perform reconciliation based on the information received in the window open sentinel **308** and the window close sentinel **330** as discussed above. More specifically, at the start of each new window, the consumer **304** may generate a state object that corresponds to the window **306**. For example, the state object may be associated with the unique identifier for the window **306**. As the consumer **304** receives additional events corresponding to the window **306** (e.g., events tagged with the window identifier), the consumer **304** may add the events to the state object.

(61) In this regard, and similar to the producer **302**, the consumer **304** might execute the reconciliation rules indicated in the window open sentinel **308** as each new event in the window **306** is received, incrementally generating the consumer-generated reconciliation attributes that will be compared to the corresponding producer-generated information in the window close sentinel **330**. In this way, the consumer **304** may be able to more quickly perform reconciliation when the window close sentinel **330** is received. Alternatively, the consumer **304** may wait to execute the reconciliation rules until after the window close sentinel **330** is received, and then do so all at once. Various combinations of the above are also possible.

(62) In some implementations, due to the reordering of events that may occur during intermediate transit of the unbounded stream, the consumer **304** might receive an event that is tagged with a window identifier for which the consumer **304** has not yet received a window open sentinel. In this situation, the consumer **304** might determine that a new window has started and create a new state object with the associated window identifier, in anticipation of receiving the window open sentinel. In this regard, the consumer **304** may be unable to begin executing the reconciliation rules until they are received in the window open sentinel.

(63) Similarly, the consumer **304** might also receive a window close sentinel before all events in the window have been received. For instance, the window close sentinel may include a producer-generated reconciliation attribute that indicates 100 events within the window, but the consumer **304** have only received **98**. To account for these situations, the consumer **304** may be configured to

include a tolerance, or expiry period, following the receipt of a window close sentinel to account for late arriving events. For instance, if a window is incomplete when the window close sentinel is received, the consumer **304** might wait an additional 10 seconds to allow the missing events to arrive before the consumer **304** determines that reconciliation of the window was unsuccessful.

(64) In this regard, the consumer **304** may notify the producer **302** whether reconciliation of each window (and shard) was successful or unsuccessful. For instance, if the consumer **304** determines that the consumer-generated reconciliation attributes for the window **306** match the producer-generated reconciliation attributes for the window **306**, the consumer **304** may notify the producer **302** that the reconciliation of the window **306** was successful. Conversely, if the consumer **304** determines that the consumer-generated and the producer-generated reconciliation attributes for the window **306** do not match, the consumer **304** may notify the producer **302** that reconciliation of the window **306** was unsuccessful, and that the events of the window should be replayed. This type of notification may be referred to herein as a replay event.

(65) Turning to the shard **310** and the shard **320** shown in FIG. 3, the producer **302** may generally follow the same framework discussed above with respect to window **306** for generating and injecting shard open and shard close sentinels into the unbounded stream of events.

(66) For example, shard **310** includes a shard open sentinel **312**, which may include an identifier associated with shard **310**. For instance, each shard within the window **306** may be assigned a shard identifier. In some cases, the shard identifier may take the form of an integer value that increases for each new shard within the window, although various other possibilities exist. Each event produced within the shard may be associated with the shard identifier (e.g., by tagging the event with the shard identifier). Accordingly, each event produced within the shard may include an indication of both a window identifier and a shard identifier. Further, the consumer **304** may use the shard identifier to sub-divide events that the consumer **304** stores within the window state object as they are received within the window **306**.

(67) Further, the shard open sentinel **312** may include an indication of one or more reconciliation rules for the shard **310**. The reconciliation rules for the shard **310** may generally take the same form as the reconciliation rules discussed above with respect to the window **306**. As above, the producer **302** may execute the shard reconciliation rules and thereby generate one or more producer-generated reconciliation attributes for the shard that will be included in the shard close sentinel **318**.

(68) In this regard, the producer **302** may define rules for reconciling, and the consumer **304** may reconcile, the unbounded stream of events at both the window level and the shard level, which may provide various benefits. At a high level, dividing the window **306** into smaller shards may allow for reconciliation to occur more frequently, which may allow for reconciliation errors to be recognized more quickly. This in turn may result in fewer events that need to be replayed. In addition, the producer **302** may define the granularity with which reconciliation is to be performed at each different level. As one possibility, the producer **302** may define shard reconciliation rules that validate the integrity of specific data fields within the unbounded stream of events, whereas the window reconciliation rules may be less granular, directly only to validating the number and identity of shards and events within the window. Alternatively, the producer **302** may define shard reconciliation rules that are less granular, and thus less computationally expensive, whereas the less frequently executed window reconciliation rules may be more granular. Other examples are also possible, and may be defined based on the business, security, and/or computational needs of the producer **302**.

(69) Similar to the window close sentinel **330**, the shard close sentinel **318** may include the shard identifier associated with shard **310**, as well as the one or more producer-generated reconciliation attributes for shard **310**. Further, the producer **302** may determine when to inject the shard close sentinel **318** into the unbounded stream of events based on similar criteria as the window close sentinel. For instance, if the producer **302** is configured to produce a new window for every 100 produced events, the producer **302** may additionally be configured to sub-divide each window into

shards that include 10 events each. Further, the producer **302** may define a similar relationship between windows and shards in a time-based rather than event-based reconciliation framework. Other examples are also possible.

(70) The consumer **304** may perform reconciliation of the shard **310** substantially as discussed above with respect to window **306**, by executing the reconciliation rules for the shard **310** and comparing the resulting consumer-generated reconciliation attributes for shard **310** with the producer-generated reconciliation attributes for shard **310** contained in the shard close sentinel **318**. If reconciliation is unsuccessful, the consumer **304** may transmit a replay event for the shard **310** to the producer **302**.

(71) As mentioned above, it should be noted that although a messaging system is not shown in FIG. **3**, there may be some implementations where an intermediate messaging system might perform reconciliation of the stream of events produced by the producer **302** before the events reach the consumer **304**. In this regard, the messaging system might perform some or all of the operations discussed above with respect to the consumer **304**.

(72) In accordance with the description above, FIG. **4** illustrates an example process **400** that may be carried out by a computing system, such as producer computing system the implements the producer **302** described above with respect to FIG. **3**. Example process **400** may begin at block **402** with the producer **302** determining a start of a window (e.g., the window **306** described above with respect to FIG. **3**) in an unbounded stream of events. As discussed above, the producer **302** may determine the start of the window **306** in various ways. As one possibility, the producer **302** may determine that no window is currently open (e.g., due to a prior window closing) and thus a new window should be opened. As another possibility, the producer **302** may determine that now window is open, but also that no events are being produced. Accordingly, the producer **302** may determine the start of a new window with it determines that a new event will be produced. Other examples are also possible.

(73) Block **404** may involve, based on determining the start of the window, injecting a window open sentinel (e.g., the window open sentinel **308**) into the unbounded stream of events that indicates one or more reconciliation rules for the window. The window open sentinel **308** may take various forms and may include various information. As one possibility, the window open sentinel **308** may include an identifier that serves to identify the window **306**. As another possibility, the window open sentinel **308** may include an indication of one or more reconciliation rules for the window **306**, which may take any of the forms discussed above.

(74) Block **406** may involve determining a start of a shard (e.g., shard **310**) in the unbounded stream of events, where the shard is within the window. The producer may determine the start of the shard **310** in various ways. As one possibility, the producer may determine that no shards are currently open within the window (e.g., either due to a window open sentinel being generated, or due to a prior shard closing).

(75) Block **408** may involve, based on determining the start of the shard **310**, injecting, into the unbounded stream of events, a shard open sentinel (e.g., shard open sentinel **312**) that indicates one or more reconciliation rules for the shard **310**. The shard open sentinel **312** may take various forms. As one possibility, the shard open sentinel **312** may include an identifier that serves to identify the shard **310**. As another possibility, the shard open sentinel **312** may indicate one or more reconciliation rules for the shard **310**, which may take any of the forms discussed above.

(76) Block **410** may involve transmitting a set of events produced within the shard **310**. The producer **302** may transmit the set of events produced within the shard **310** in various ways. As one possibility, the producer **302** may transmit the set of events produced within the shard to a messaging system that is configured to facilitate delivery of the set of events produced within the shard **310** to a consumer (e.g., consumer **304**). The producer **302** may transmit the set of events produced within the shard **310** in other ways as well.

(77) Block **412** may involve executing the one or more reconciliation rules for the shard **310** for a

set of events produced within the shard (e.g., event **314** and event **316**) to thereby generate one or more producer-generated reconciliation attributes for the shard **310**. In this respect, the one or more producer-generated reconciliation attributes for the shard **310** may correspond to the one or more reconciliation rules for the shard **310**. For instance, if the one or more reconciliation rules for the shard includes a rule to count the number of events produced within the shard **310**, the resulting producer-generated reconciliation attribute may identify the number of the events in the shard **310** (e.g., 10 events). Further, the one or more producer-generated reconciliation attributes for the shard **310** may represent the expected result that the consumer **304** should obtain when executing the one or more reconciliation rules for the shard **310** when attempting reconciliation of the shard.

(78) Further, the producer **302** may execute the one or more reconciliation rules for the shard at various times. As one possibility, the producer **302** may execute the one or more reconciliation rules for the shard **310** after the entire set of events are produced within the shard **310**. As another possibility, the producer **302** may execute the one or more reconciliation rules for the shard **310** after each event in the set of events is produced within the shard **310**. The producer **302** may execute the one or more reconciliation rules for the shard **310** at other times as well.

(79) Block **414** may involve determining a close of the shard **310** in the unbounded stream of events. The producer **302** may determine the close of the shard in various ways. As one possibility, the producer **302** may determine that one or more conditions have been met, examples of which may include (i) determining that a target number of events have been produced within the shard **310** (e.g., events) or (ii) determining that a given time interval has expired (e.g., thirty seconds, one minute, one hour, etc.). The producer **302** may determine the close of the shard **310** in other ways as well.

(80) Block **416** may involve, based on determining the close of the shard **310**, injecting, into the unbounded stream of events, a shard close sentinel (e.g., shard close sentinel **318**) that indicates the one or more producer-generated reconciliation attributes for the shard **310** that were generated by the producer **302** executing the one or more reconciliation rules for the shard **310**. The shard close sentinel **318** may take various forms and may include data such as (i) the shard identifier, and (ii) the one or more producer-generated reconciliation attributes for the shard. The shard close sentinel **318** may take other forms as well.

(81) Block **418** may involve executing the one or more reconciliation rules for the window **306** for a set of events produced within the window (e.g., the events within shard **310** and shard **320**) to thereby generate one or more producer-generated reconciliation attributes for the window **306**. In this respect, the one or more producer-generated reconciliation attributes for the window **306** may correspond to the one or more reconciliation rules for the window. Further, the one or more producer-generated reconciliation attributes for the window may represent the expected result that consumer **304** should obtain when executing the one or more reconciliation rules for the window **306** when attempting reconciliation of the window **306**.

(82) Further, the producer **302** may execute the one or more reconciliation rules for the window **306** at various times. As one possibility, the producer **302** may execute the one or more reconciliation rules for the window **306** after the entire set of events are produced within the window **306**. As another possibility, the producer **302** may execute the one or more reconciliation rules for the window **306** after each event in the set of events is produced within the window **306**. The producer **302** may execute the one or more reconciliation rules for the window **306** at other times as well.

(83) Block **420** may involve determining a close of the window **306** in the unbounded stream of events. The producer **302** may determine the close of the window **306** in various ways. As one possibility, the producer **302** may determine that one or more conditions have been met, examples of which may include (i) determining that a target number of events have been published within the window **306**, (ii) determining that a target number of shards have been opened and closed within the window, or (iii) determining that a given time interval has expired. The producer **302** may

determine the close of the window **306** in other ways as well.

(84) Block **422** may involve, based on determining the close of the window **306**, injecting, into the unbounded stream of events, a window close sentinel (e.g., window close sentinel **330**) that indicates the one or more producer-generated reconciliation attributes for the window **306**. The window close sentinel **330** may take various forms and may include data such as (i) the window identifier, and (ii) the one or more producer-generated reconciliation attributes for the window. The window close sentinel **330** may take other forms as well.

(85) It should be understood that while the producer **302** is performing the functions above, one or more consumers **304** may be performing reconciliation on the window(s) or shard(s). In turn, if reconciliation of a given window or shard fails, the producer **302** may receive a request to resend the set of events produced within a given window or shard. In response to receiving such a request, the producer may “replay” the window or shard identified in the request by repeating one or more of the functions described above.

(86) For example, a producer **302** may inject a shard open sentinel and a shard close sentinel in accordance with the description above. At some point in time after the producer **302** has injected the shard close sentinel (e.g., before or after the producer **302** has injected the window close sentinel), the producer **302** may receive an indication that reconciliation of the shard was unsuccessful. In response to receiving the indication, the producer **302** may replay the shard by reproducing the set of events for the shard (e.g., by retransmitting the events to a messaging system). In turn, one or more downstream consumer can reattempt reconciliation of the shard.

(87) In some circumstances, the producer **302** may, in addition to replaying the shard, replay the entire window of which the replayed shard is a part. In this respect, the producer **302** may replay the window by reproducing the set of events for the entire window. In turn, a downstream consumer can reattempt reconciliation of the entire window of which the shard is a part.

(88) In accordance with the description above, FIG. 5 illustrates an example process **500** that may be carried out by a computing system, such as consumer computing system the implements the consumer **304** described above with respect to FIG. 3. Example process **500** may begin at block **502** with the consumer **304** detecting a window open sentinel (e.g., window open sentinel **308**) for a window (e.g., window **306** described above with respect to FIG. 3) within an unbounded stream of events, where the window open sentinel **308** indicates one or more reconciliation rules for the window **306**. The consumer **304** may detect the window open sentinel **308** for the window **306** as a result of monitoring the unbounded stream of events for a window open sentinel, among other possibilities. The window open sentinel **308** may indicate a window identifier as well as one or more reconciliation rules for the window, which may take the form of any of the reconciliation rules discussed above.

(89) Block **504** may involve detecting a shard open sentinel (e.g., shard open sentinel **312**) for a shard (e.g., shard **310**) within the window **306**, where the shard open sentinel **312** indicates one or more reconciliation rules for the shard **310**. As above, the reconciliation rules for the shard **310** may take the form of any of the reconciliation rules discussed above.

(90) Block **506** may involve receiving a set of events produced within the shard **310**. The consumer **304** may receive the set of events produced within the shard in various ways. As one possibility, the consumer may receive the set of events from a messaging system configured to facilitate the delivery of the set events produced within the shard **310** from the producer **302** to the consumer **304**. In addition, the consumer **304** may store each event that is received within the shard in a state object that the consumer **304** created for the window **306**.

(91) Block **508** may involve detecting a shard close sentinel (e.g., shard close sentinel **318**) for the shard **310**, where the shard close sentinel **318** includes an indication of one or more producer-generated reconciliation attributes for the shard **310**. For example, the consumer **304** may be configured to monitor the unbounded stream of events for the shard close sentinel **318**. The consumer **304** may detect the shard close sentinel **318** in other ways as well.

(92) Block **510** may involve executing the one or more reconciliation rules for the shard **310** for a set of events produced within the shard **310** to thereby generate one or more consumer-generated reconciliation attributes for the shard **310**. In this respect, the one or more consumer-generated reconciliation attributes for the shard **310** may correspond to the one or more reconciliation rules for the shard **310**. For instance, if the one or more reconciliation rules for the shard includes a rule to count the number of events produced within the shard **310**, the corresponding consumer-generated reconciliation attribute may identify the number of the events in the shard **310**. Once the one or more consumer-generated reconciliation attributes for the shard are generated, the consumer **304** may store each of the one or more consumer-generated reconciliation attributes for each event produced within the shard **310** in the data object for the shard **310**.

(93) Block **512** may involve, based on detecting the shard close sentinel **318**, reconciling the determined one or more consumer-generated reconciliation attributes for the shard **310** against the one or more producer-generated reconciliation attributes for the shard **310**. For example, the consumer **304** may compare the determined one or more consumer-generated reconciliation attributes for the shard **310** with the one or more producer-generated reconciliation attributes for the shard **310**. In this respect, if the determined one or more consumer-generated reconciliation attributes for the shard **310** are different than the one or more producer-generated reconciliation attributes for the shard **310**, the consumer **304** may determine that reconciliation of the shard was unsuccessful. Conversely, if the consumer **304** determines that the one or more consumer-generated reconciliation attributes for the shard **310** are the same as the one or more producer-generated reconciliation attributes for the shard **310**, the consumer **304** may determine that the reconciliation of the shard **310** was successful.

(94) Block **514** may involve detecting a window close sentinel (e.g., the window close sentinel **330**), wherein the window close sentinel **330** includes an indication of one or more producer-generated reconciliation attributes for the window **306**. For example, the consumer **304** may be configured to monitor the unbounded stream of events for the window close sentinel **330**. The consumer **304** may detect the window close sentinel **330** in other ways as well.

(95) Block **516** may involve executing the one or more reconciliation rules for the window **306** for a set of events produced within the window **306** to thereby generate one or more consumer-generated reconciliation attributes for the window **306**. In this respect, the one or more consumer-generated reconciliation attributes for the window **306** may correspond to the one or more reconciliation rules for the window **306**. For instance, if the one or more reconciliation rules for the window includes a rule to count the number of shards within the window **306**, there may be a corresponding consumer-generated reconciliation attribute that identifies the number of the shards in the window **306**. Once the one or more consumer-generated reconciliation attributes for the window **306** are generated, the consumer **304** may store the one or more consumer-generated reconciliation attributes for each event produced within the window in the state object for the window **306**.

(96) Block **518** may involve, based on detecting the window close sentinel **330**, reconciling the determined one or more consumer-generated reconciliation attributes for the window **306** against the one or more producer-generated reconciliation attributes for the window **306**. For example, the consumer **304** may compare the determined one or more consumer-generated reconciliation attributes for the window **306** with the one or more producer-generated reconciliation attributes for the window **306** to determine whether reconciliation of the window **306** is successful, as discussed above.

(97) For the example process **400** of FIG. **4**, the example processes **500** shown in FIG. **5**, and other processes described herein, the flowcharts show functionality and operation of one possible implementation of embodiments disclosed herein. In this regard, each block may represent a module, a segment, or a portion of program code, which includes one or more instructions executable by a processor for implementing specific logical functions or steps in the process. The

program code may be stored on any type of computer readable medium, for example, such as a storage device including a disk or hard drive. The computer readable medium may include non-transitory computer readable medium, for example, such as computer-readable media that stores data for short periods of time like register memory, processor cache and Random Access Memory (RAM). The computer readable medium may also include non-transitory media, such as secondary or persistent long term storage, like read only memory (ROM), optical or magnetic disks, compact-disc read only memory (CD-ROM), for example. The computer readable media may also be any other volatile or non-volatile storage systems. The computer readable medium may be considered a computer readable storage medium, for example, or a tangible storage device. In addition, for the example processes **400** and **500**, each block shown in FIG. **4** and FIG. **5** may represent circuitry that is wired to perform the specific logical functions in the process.

(98) Turning next to FIG. **6**, a simplified block diagram is provided to illustrate some structural components that may be included in an example computing system **600**. For example, computing system **600** may implement one or more of the software components shown in FIG. **2** and FIG. **3** (e.g., producer **302** or consumer **304**) and may be configured to carry out any of the various functions disclosed herein—including but not limited to the functions included in the example data flows described with reference to FIGS. **4-5**. At a high level, computing system **600** may generally comprise any one or more computer systems (e.g., one or more servers) that collectively include at least a processor **602**, data storage **604**, a communication interface **606**, and a user interface **608**, all of which may be communicatively linked by a communication link **610** that may take the form of a system bus, a communication network such as a public, private, or hybrid cloud, or some other connection mechanism. Each of these components may take various forms.

(99) Processor **602** may comprise one or more processing components, such as general-purpose processors (e.g., a single- or a multi-core CPU), special-purpose processors (e.g., a GPU, application-specific integrated circuit, or digital-signal processor), programmable logic devices (e.g., a field programmable gate array), controllers (e.g., microcontrollers), and/or any other processor components now known or later developed. In line with the discussion above, it should also be understood that processor **602** could comprise processing components that are distributed across a plurality of physical computing devices connected via a network, such as a computing cluster of a public, private, or hybrid cloud.

(100) In turn, data storage **604** may comprise one or more non-transitory computer-readable storage mediums that are collectively configured to store (i) program instructions that are executable by processor **602** such that the computing system **600** is configured to perform certain functions, and (ii) data that may be received, derived, or otherwise stored, for example, in one or more databases, file systems, repositories, or the like, by the computing system **600**. In this respect, the one or more non-transitory computer-readable storage mediums of data storage **604** may take various forms, examples of which may include volatile storage mediums such as random-access memory, registers, cache, etc. and non-volatile storage mediums such as read-only memory, a hard-disk drive, a solid-state drive, flash memory, an optical-storage device, etc. In line with the discussion above, it should also be understood that data storage **604** may comprise computer-readable storage mediums that are distributed across a plurality of physical computing devices connected via a network, such as a storage cluster of a public, private, or hybrid cloud that operates according to technologies such as AWS for Elastic Compute Cloud, Simple Storage Service, etc.

(101) Communication interface **606** may take the form of any one or more interfaces that facilitate wireless and/or wired communication between computing system **600** and other systems or devices. In this respect, each such interface may take any of various forms, examples of which may include an Ethernet interface, a serial bus interface (e.g., Firewire, USB 3.0, etc.), a chipset and antenna adapted to facilitate wireless communication, and/or any other interface that provides for any of various types of wireless communication (e.g., Wi-Fi communication, cellular communication, short-range wireless protocols, etc.) and/or wired communication. Other

configurations are possible as well.

(102) The computing system **600** may additionally include or have an interface for connecting to one or more user-interface components that facilitate user interaction with the computing system **600**, such as a keyboard, a mouse, a trackpad, a display screen, a touch-sensitive interface, a stylus, a virtual-reality headset, and/or one or more speaker components, among other possibilities.

(103) It should be understood that the computing system **600** is one example of a computing system that may be used with the embodiments described herein. Numerous other arrangements are possible and contemplated herein. For instance, in other embodiments, the computing system **600** may include additional components not pictured and/or more or fewer of the pictured components.

CONCLUSION

(104) This disclosure makes reference to the accompanying figures and several example embodiments of the disclosed innovations that have been described above. One of ordinary skill in the art should understand that such references are for the purpose of explanation only and are therefore not meant to be limiting. Part or all of the disclosed systems, devices, and methods may be rearranged, combined, added to, and/or removed in a variety of manners without departing from the true scope and spirit of the present invention, which will be defined by the claims.

(105) Further, to the extent that examples described herein involve operations performed or initiated by actors, such as “humans,” “curators,” “users” or other entities, this is for purposes of example and explanation only. The claims should not be construed as requiring action by such actors unless explicitly recited in the claim language.

Claims

1. A computing system comprising: a network interface for communicating over at least one data network; at least one processor; at least one non-transitory computer-readable medium; and program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to: detect a window open sentinel for a window within an unbounded stream of events, wherein the window open sentinel indicates one or more reconciliation rules for the window; detect a shard open sentinel for a shard within the window, wherein the shard open sentinel indicates one or more reconciliation rules for the shard; receive a set of events produced within the shard; detect a shard close sentinel for the shard, wherein the shard close sentinel includes an indication of one or more producer-generated reconciliation attributes for the shard; execute the one or more reconciliation rules for the shard for a set of events produced within the shard to thereby generate one or more consumer-generated reconciliation attributes for the shard; based on detecting the shard close sentinel, reconcile the one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard; detect a window close sentinel, wherein the window close sentinel includes an indication of one or more producer-generated reconciliation attributes for the window; execute the one or more reconciliation rules for the window for a set of events produced within the window to thereby generate one or more consumer-generated reconciliation attributes for the window; and based on detecting the window close sentinel, reconcile the one or more consumer-generated reconciliation attributes for the window against the one or more producer-generated reconciliation attributes for the window.

2. The computing system of claim 1, wherein the computing system is a consumer computing system, and wherein the program instructions that are executable by the at least one processor such that the consumer computing system is configured to reconcile the one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard comprise program instructions that are executable by the at least one processor such that the consumer computing system is configured to: determine that the one or more consumer-generated reconciliation attributes for the shard match the one or more

producer-generated reconciliation attributes for the shard; and transmit, to a producer computing system, an indication that the shard was successfully reconciled.

3. The computing system of claim 1, wherein the computing system is a consumer computing system, and wherein the program instructions that are executable by the at least one processor such that the consumer computing system is configured to reconcile the one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard comprise program instructions that are executable by the at least one processor such that the consumer computing system is configured to: determine that the one or more consumer-generated reconciliation attributes for the shard does not match the one or more producer-generated reconciliation attributes for the shard; discard the set of events produced within the shard; and transmit, to a producer computing system, a request to retransmit the set of events produced within the shard.

4. The computing system of claim 1, further comprising program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to: create a data object corresponding to the shard; and after each event in the set of events produced within the shard is received, add the event to the data object for the shard.

5. The computing system of claim 4, wherein the program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to receive the set of events produced within the shard comprise program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to receive at least one event in the set of events produced within the shard before detecting the shard open sentinel; and wherein the program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to create the data object corresponding to the shard comprise program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to create the data object corresponding to the shard based on receiving the at least one event in the set of events produced within the shard before detecting the shard open sentinel.

6. The computing system of claim 1, wherein the program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to execute the one or more reconciliation rules for the shard comprise program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to execute the one or more reconciliation rules for the shard after all events in the set of events produced within the shard are received.

7. The computing system of claim 1, wherein the program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to execute the one or more reconciliation rules for the shard comprise program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing system is configured to execute the one or more reconciliation rules for the shard after each event in the set of events produced within the shard is received.

8. The computing system of claim 1, wherein the window open sentinel includes an indication of a window identifier, and wherein each event in the set of events received within the window includes an indication of the window identifier.

9. The computing system of claim 1, wherein the one or more reconciliation rules for the window comprise instructions for determining (i) a given number of events produced within the window, (ii) a given number of shards produced within the window, and (iii) a signature for each event

produced within the window.

10. The computing system of claim 1, wherein the one or more reconciliation rules for the window comprise instructions for determining one or more of (i) a summation of a given event attribute across all events produced within the window, (ii) a number of a given type of event produced within the window, or (iii) a signature for each of a given type of event produced within the window.

11. The computing system of claim 1, wherein the one or more reconciliation rules for the shard comprise instructions for determining (i) a given number of events produced within the shard and (ii) a signature for each event produced within the shard.

12. The computing system of claim 1, wherein the one or more reconciliation rules for the shard comprise instructions for determining one or more of (i) a summation of a given event attribute across all events produced within the shard, (ii) a number of a given type of event produced within the shard, or (iii) a signature for each of a given type of event produced within the shard.

13. The computing system of claim 1, wherein the one or more reconciliation rules for the window are different from the one or more reconciliation rules for the shard.

14. A non-transitory computer-readable medium, wherein the non-transitory computer-readable medium is provisioned with program instructions that, when executed by at least one processor, cause a computing system to: detect a window open sentinel for a window within an unbounded stream of events, wherein the window open sentinel indicates one or more reconciliation rules for the window; detect a shard open sentinel for a shard within the window, wherein the shard open sentinel indicates one or more reconciliation rules for the shard; receive a set of events produced within the shard; detect a shard close sentinel for the shard, wherein the shard close sentinel includes an indication of one or more producer-generated reconciliation attributes for the shard; execute the one or more reconciliation rules for the shard for a set of events produced within the shard to thereby generate one or more consumer-generated reconciliation attributes for the shard; based on detecting the shard close sentinel, reconcile the one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard; detect a window close sentinel, wherein the window close sentinel includes an indication of one or more producer-generated reconciliation attributes for the window; execute the one or more reconciliation rules for the window for a set of events produced within the window to thereby generate one or more consumer-generated reconciliation attributes for the window; and based on detecting the window close sentinel, reconcile the one or more consumer-generated reconciliation attributes for the window against the one or more producer-generated reconciliation attributes for the window.

15. The non-transitory computer-readable medium of claim 14, wherein the computing system is a consumer computing system, and wherein the program instructions that are executable by the at least one processor such that the consumer computing system is configured to reconcile the one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard comprise program instructions that are executable by the at least one processor such that the consumer computing system is configured to: determine that the one or more consumer-generated reconciliation attributes for the shard match the one or more producer-generated reconciliation attributes for the shard; and transmit, to a producer computing system, an indication that the shard was successfully reconciled.

16. The non-transitory computer-readable medium of claim 14, wherein the computing system is a consumer computing system, and wherein the program instructions that are executable by the at least one processor such that the consumer computing system is configured to reconcile the one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard comprise program instructions that are executable by the at least one processor such that the consumer computing system is configured to: determine that the one or more consumer-generated reconciliation attributes for the shard does not match the

one or more producer-generated reconciliation attributes for the shard; discard the set of events produced within the shard; and transmit, to a producer computing system, a request to retransmit the set of events produced within the shard.

17. The non-transitory computer-readable medium of claim 14, wherein the non-transitory computer-readable medium is also provisioned with program instructions that, when executed by at least one processor, cause the computing system to: create a data object corresponding to the shard; and after each event in the set of events produced within the shard is received, add the event to the data object for the shard.

18. The non-transitory computer-readable medium of claim 17, wherein the program instructions that, when executed by at least one processor, cause the computing system to receive the set of events produced within the shard comprise program instructions that, when executed by at least one processor, cause the computing system to receive at least one event in the set of events produced within the shard before detecting the shard open sentinel; and wherein the program instructions that, when executed by at least one processor, cause the computing system to create the data object corresponding to the shard comprise program instructions that, when executed by at least one processor, cause the computing system to create the data object corresponding to the shard based on receiving the at least one event in the set of events produced within the shard before detecting the shard open sentinel.

19. The non-transitory computer-readable medium of claim 14, wherein the one or more reconciliation rules for the window are different from the one or more reconciliation rules for the shard.

20. A non-transitory computer-readable medium, wherein the non-transitory computer-readable medium is provisioned with program instructions that, when executed by at least one processor, cause a computing system to: detect a window open sentinel for a window within an unbounded stream of events, wherein the window open sentinel indicates one or more reconciliation rules for the window; detect a shard open sentinel for a shard within the window, wherein the shard open sentinel indicates one or more reconciliation rules for the shard; receive a set of events produced within the shard; detect a shard close sentinel for the shard, wherein the shard close sentinel includes an indication of one or more producer-generated reconciliation attributes for the shard; execute the one or more reconciliation rules for the shard for a set of events produced within the shard to thereby generate one or more consumer-generated reconciliation attributes for the shard; based on detecting the shard close sentinel, reconcile the one or more consumer-generated reconciliation attributes for the shard against the one or more producer-generated reconciliation attributes for the shard; detect a window close sentinel, wherein the window close sentinel includes an indication of one or more producer-generated reconciliation attributes for the window; execute the one or more reconciliation rules for the window for a set of events produced within the window to thereby generate one or more consumer-generated reconciliation attributes for the window; and based on detecting the window close sentinel, reconcile the one or more consumer-generated reconciliation attributes for the window against the one or more producer-generated reconciliation attributes for the window.
